

Global Extreme Weather and Early Childhood Undernourishment

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Abstract

Weather extremes in early life may affect child growth and nutritional status. Global evidence on these effects across developing countries remains limited. This paper examines how local deviations in temperature and precipitation affect early-childhood growth outcomes in the first 1,000 days of development – from conception to 24 months old – when children are most developmentally vulnerable. We exploit Demographic and Health Surveys data linked to high-resolution ERA5 weather data to reconstruct the full daily weather exposure history of over 1.2 million children from 57 low- and middle-income countries. Identification compares children surveyed in the same province-climate zone and month, while accounting for time-varying country-level factors, who experienced different weather realizations *in utero* and early in life. We find that weather exposure has nonlinear effects on child growth status through distinct periods of vulnerability over early life. *In utero*, drought exposure and extreme cold exposure both persistently lower height-for-age among children ages 3–5 years. We observe similar drought effects shortly after birth, while temperature effects are muted. By age one-year-old, extreme rainfall exposure dominates impacts. These results suggest that early-life weather shocks can have effects on child growth status that persist through older ages, with implications for child development in vulnerable settings.

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1 Introduction

Climate change is already affecting weather extremes across regions (Pörtner et al., 2022). These impacts disproportionately affect vulnerable populations, particularly in low- and middle-income countries (Hallegatte, 2016; Carleton et al., 2022). In these settings, households face both high exposure to climate shocks (Pörtner et al., 2022) and limited capacity to adapt (Carleton et al., 2022; Kahn, 2016). Economic activity relies heavily on climate-sensitive sectors such as agriculture, making food availability and prices highly responsive to weather fluctuations (Hultgren et al., 2025; Nath, 2022; World Bank, 2019; Heinemann et al., 2011; Dewey, 2016). At the same time, weak health systems, limited infrastructure, and incomplete social safety nets constrain households’ ability to mitigate the effects of these shocks. As a result, identical weather realizations can generate larger losses in developing countries than in high-income economies.

These risks are particularly salient for children, who are highly sensitive to environmental shocks and have limited capacity to adapt (Roth et al., 2017). Exposure to extreme weather during the prenatal period and early childhood affects health through multiple channels, including changes in food availability, disease exposure, and maternal health (Bhutta et al., 2013, 2023; Ramakrishnan et al., 2012). A growing literature examines the impacts of climate and weather on agricultural production and population health (Hultgren et al., 2025; Carleton et al., 2022). However, less is known about how weather fluctuations translate into child-level growth outcomes.

We study how extreme weather affects child stunting using height-for-age z-scores in low- and middle-income countries. We use child-level anthropometric data from the Demographic and Health Surveys (DHS), covering over 1.2 million children in 58 low- and middle-income countries between 1990 and 2023. We link these data to high-resolution ERA5 weather data based on household GPS coordinates and interview timing. For each child, we reconstruct the full history of daily weather exposure from conception to birth to development until the survey date. Our identification comes from variation in children’s weather exposure histories, after controlling for country-year and province-climate zone-month fixed effects. Country-year fixed effects control for national factors that change across years such as macroeconomic conditions, policies, and aggregate shocks, while province-climate zone-month fixed effects absorb location-specific seasonal patterns in weather and

health outcomes. The remaining variation is driven by idiosyncratic temperature and precipitation shocks relative to typical monthly patterns within each province-climate zone over the lifetime of a given child.

We further exploit differences in the timing of these shocks across children’s early-life exposure histories. For example, a drought occurring in a given location and time may affect children differently depending on their developmental stage at the time of the shock: some children may have experienced the drought during gestation, others during infancy, and others at later stages of development. We leverage this variation to estimate how the effects of temperature and precipitation shocks differ across the early-life cycle. We find extreme weather reduce height-for-age (HAZ) among children older than 36 months, but the effects vary by both timing of exposure and the type of shock. Cold exposure lowers HAZ primarily in utero, with the largest effects concentrated in late pregnancy. Drought lowers HAZ before birth, whereas heavy rainfall decreases HAZ after birth. These results indicate that the lasting effects of early-life weather exposure depend on both developmental timing and the type of shock.

This study focuses on z-scores in height-for-age. HAZ, in turn, is the measure used to define stunting, a key developmental outcome with potentially important welfare implications. Early childhood stunting has been linked to lower earnings and negative economic outcomes later in life ([Victora et al., 2008](#); [Hoddinott et al., 2013](#)). Moreover, stunting that remains after age two is less likely to be fully reversed and may therefore persist into later life ([Soliman et al., 2021](#); [Dewey and Begum, 2011](#)). Thus, while DHS data is only compiled for children under 5 years of age, our outcome sheds light on developmental impacts that persist into adulthood and have been linked to important welfare effects through long-term earnings.

This study makes three main contributions to the literature. First, we provide micro-level evidence from a large set of countries by linking harmonized DHS data from 58 low- and middle-income countries to high-resolution weather data. Most existing evidence on the effects of weather on child health is based on single-country or regional analyses, limiting external validity ([Baker and Anttila-Hughes, 2020](#); [Blom et al., 2019](#); [Hoddinott et al., 2013](#); [Carlton et al., 2014](#)). Our broader scope allows us to assess whether these relationships generalize across diverse climatic and economic contexts.

Second, we focus on child stunting to directly measure the effects of weather shocks on early life growth.

Previous research has identified potential pathways through which weather affects childhood nutrition. These pathways may operate through impacts on agricultural productivity and food prices (Hultgren et al., 2025; Lobell et al., 2011; Ray et al., 2015; Lesk et al., 2016), as well as increased disease risk through compromised sanitation systems (Levy et al., 2016; Cann et al., 2013; Carlton et al., 2014) and vector-borne disease transmission (Patz and Reisen, 2001; Mordecai et al., 2019). However, their implications for child development rely on intermediate outcomes, such as crop yields or disease incidence, rather than directly measuring developmental status. Instead, we use standardized anthropometric measures to capture the impact of weather on early childhood outcomes, allowing for consistent comparisons across countries.

Third, we leverage the full history of weather exposure from the prenatal period through early childhood to examine how the timing of shocks shapes child growth and nutritional outcomes. Child growth trajectories are particularly sensitive during the first 1,000 days (Victora et al., 2010; De Onis and Branca, 2016), when exposure to adverse conditions can have lasting developmental effects (Victora et al., 2008; Jensen et al., 2017). Nutritional deficits incurred during these periods may be difficult to reverse, as catch-up growth becomes increasingly limited after early childhood (Blom et al., 2019; De Onis and Branca, 2016). As a result, early life shocks can have persistent effects on a child’s growth status and may have implications for long-run human capital (Hoddinott et al., 2013; Alderman et al., 2006; Gluckman et al., 2007). By observing the complete exposure profile for each child, we can identify differential sensitivity to weather shocks across developmental stages and assess the persistence of early life impacts.

Our work is most closely related to Baker and Anttila-Hughes (2020). They examined temperature effects on child anthropometric outcomes in sub-Saharan Africa, and found that higher temperatures worsen acute child malnutrition over the 12 months prior to the survey. Our results are broadly consistent with this pattern. We study a broader set of countries, incorporate both temperature and precipitation, and measure exposure over a child’s full lifetime, spanning the prenatal period and early childhood – a much longer exposure window than previous studies.

The remainder of this paper is organized as follows. Section 2 describes the data, including DHS anthropometric measures and ERA5 weather data, and explains the geospatial matching procedure. Section

3 presents the empirical strategy using flexible functional forms to capture nonlinear weather effects. Section 4 presents the main results on temperature and precipitation impacts on child health outcomes over the life cycle and examines heterogeneity by child sex and household wealth. Section 5 concludes with policy implications.

2 Data

In this study, we employ publicly available DHS survey data to study child health outcomes in developing countries. We combine child-level anthropometric data with high-resolution ERA5 weather data by leveraging information on household GPS coordinates, child age, and survey timing. This allows us to recreate the entire history of the child’s daily weather exposure from conception through early childhood up to the survey date. Our primary outcome is stunting, measured using stunting(height-for-age z-scores) at the time of the survey. The following subsections describe each data source and explain how we link them geographically and temporally.

2.1 Demographic and Health Surveys (DHS)

Our primary data source is the Demographic and Health Surveys (DHS), which provide nationally representative and comparable household data for participating low- and middle-income countries. We use child health data from 57 countries with available georeferenced information, covering approximately 44% of the global population.¹ Our dataset includes over 1.2 million observations of children aged under five years from 1990 to 2024.

The DHS collects child-level anthropometric information, including age, sex, height, and weight measurements. It also reports standardized nutrition indicators based on WHO Child Growth Standards ([World Health Organization, 2006](#)). We use height-for-age(HAZ), weight-for-age(WAZ), and weight-for-height(WHZ) Z-scores provided by DHS. the Z-scores measure how many standard deviations a child’s measurement

¹Our sample is restricted to countries with child health module and GPS data necessary to link households to local weather exposure. While the DHS program spans more than 90 countries, this subset reflects data availability and does not represent all low- and middle-income countries.

deviates from the reference median for the same age and sex. DHS computes Z-scores using WHO reference distributions, ensuring consistency across countries and survey rounds. We construct binary indicators of nutritional status from these Z-scores: stunting (low height-for-age), underweight (low weight-for-age), and wasting (low weight-for-height). For each measure, children are classified as moderately malnourished if the corresponding Z-score lies between -3 and -2 , and as severely malnourished if it falls below -3 .

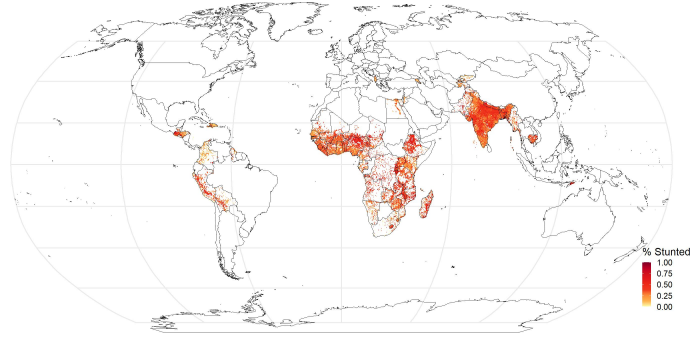
Stunting represents cumulative nutritional deficits resulting from prolonged inadequate nutrition and repeated infections (WHO, 2024). The effects of stunting are difficult to reverse and result in lasting consequences for adult height, cognitive development, and economic outcomes (Victora et al., 2008; Hoddinott et al., 2013). Wasting reflects acute malnutrition due to recent disruptions in food availability, disease exposure, or caregiving practices (WHO, 2024). Underweight serves as a composite indicator combining both chronic and acute nutritional deficits (WHO, 2024). These anthropometric outcomes directly capture the effects of both pathways through which weather affects child development.

On average in our sample, 34% of children are stunted, compared to 22% underweight and 11% wasted. This pattern aligns with prior evidence that chronic growth deficits are more widespread and persistent than acute weight loss in developing countries. The lower prevalence of wasting reflects its transitory character and sensitivity to immediate nutritional shocks. Underweight exhibits intermediate status between stunting and wasting.

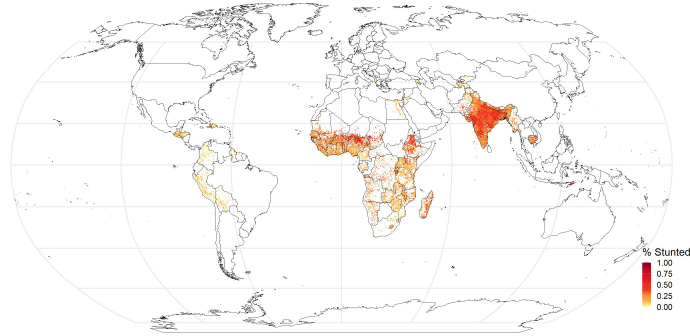
Figure 1 summarizes the geographic distribution of malnutrition prevalence across our 57 country sample. The high spatial resolution reveals meaningful within-country differences in child outcomes that national averages do not capture. Panel (a) shows that stunting varies widely across and within countries, with pronounced patterns in Sub-Saharan Africa and South Asia. Panel (b) indicates underweight follows a similar spatial distribution but at lower overall prevalence. Panel (c) shows that wasting is less prevalent but remains unevenly distributed across subnational regions.

Figure 1: Sub-National Child Malnutrition Prevalence Across 57 Countries

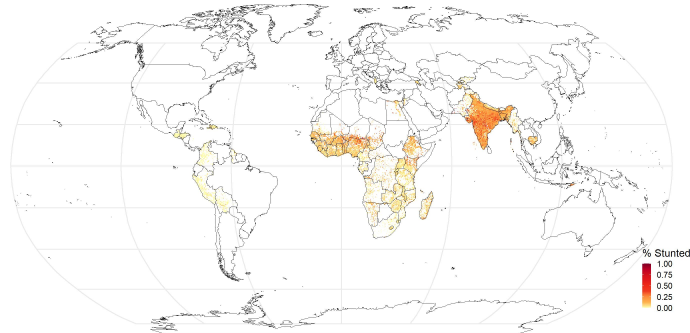
(a) Stunting



(b) Underweight



(c) Wasting



Notes: The maps display the prevalence rates of moderate and severe ($Z\text{-score} < -2$) malnutrition at the ERA5 grid cell level ($0.25^\circ \times 0.25^\circ$ resolution). Each pixel represents the mean malnutrition rate among DHS children under five located within that grid cell across all survey rounds. The color scale ranges from 0 (yellow) to 1 (dark red).

2.2 ERA5 Weather Data

We construct weather variables using the ERA5 Hourly Data on Single Levels from 1940 to Present, produced by the European Centre for Medium-Range Weather Forecasts (ECMWF) ([Hersbach et al., 2023](#)). ERA5 provides global hourly estimates at high resolution ($0.25^\circ \times 0.25^\circ$ spatial resolution, approximately $27 \text{ km} \times 27 \text{ km}$ near the equator), offering comprehensive coverage and temporal consistency. Our primary variables are air temperature and precipitation. Temperature data are the temperature of the air at 2 m above the surface of land. Precipitation is accumulated liquid and frozen water, consisting of rain and snow that falls on the surface of earth. We aggregate these hourly data into daily means for temperature ($^\circ\text{F}$) and monthly totals for precipitation (mm) at the grid cell level.

Figure 2 maps the geographic distribution of extreme weather exposure. This highlights the fact that extreme temperature and precipitation conditions are common across many regions. Extreme heat exposure (above 90°F) is concentrated in northern and western Africa, the Middle East, and South Asia, whereas extreme cold exposure (below 50°F) occurs in mountainous and higher-latitude regions. Similarly, drought conditions (below 5mm) are concentrated in the Sahel, the Middle East, and parts of southern Africa. Heavy rainfall (above 300 mm) is observed in equatorial Africa, the South Asian monsoon zones, and tropical regions. Regions experiencing both extremes (purple) indicate exposure to multiple weather risks throughout the year. Many of these high-exposure regions are located in low- and middle-income countries.

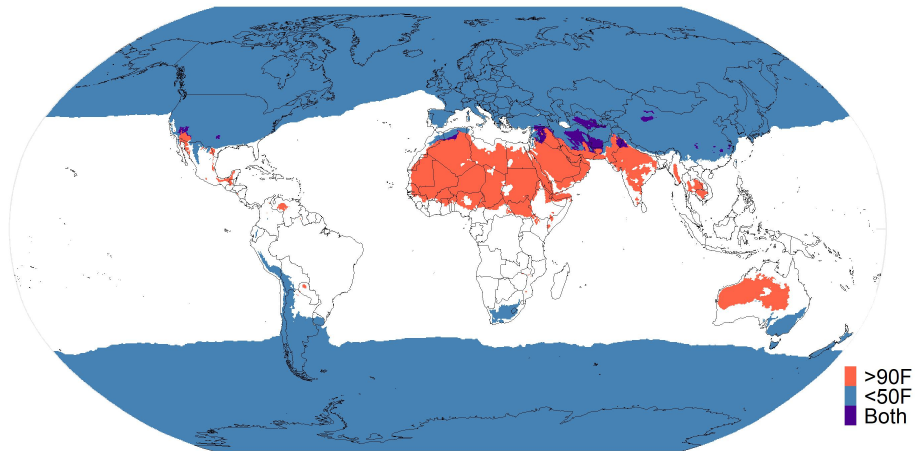
2.3 Geospatial Matching of ERA5 Data with DHS GPS Locations

To construct child-level measures of weather exposure, we link ERA5 gridded weather data to DHS household GPS points based on geolocation and interview timing. The DHS provides GPS coordinates for each household. For confidentiality protection, the DHS randomly displaces GPS coordinates by up to 2 km in urban areas and up to 5 km (occasionally 10 km) in rural areas. We use this data to match to the corresponding ERA5 grid cell.

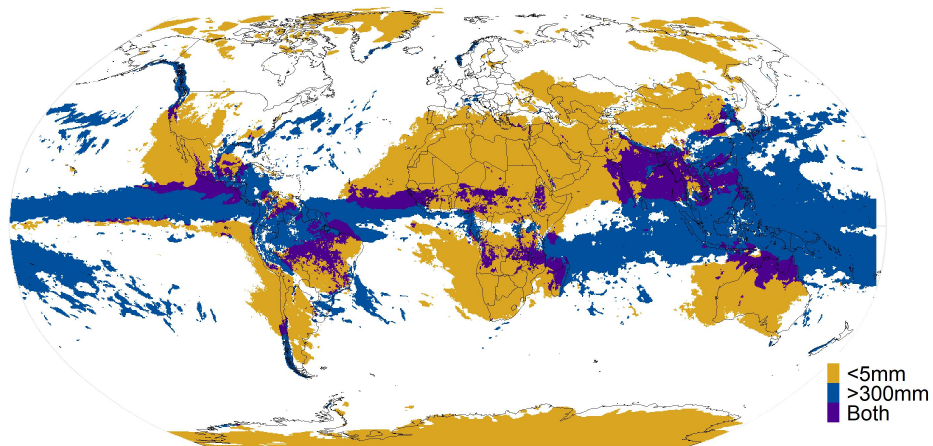
For each child, we assign local weather exposures from the prenatal period through the month of survey at the corresponding GPS location. Temperature is measured using daily averages ($^\circ\text{F}$), while precipitation is

Figure 2: Geographic Distribution of Weather Extremes

(a) Temperature

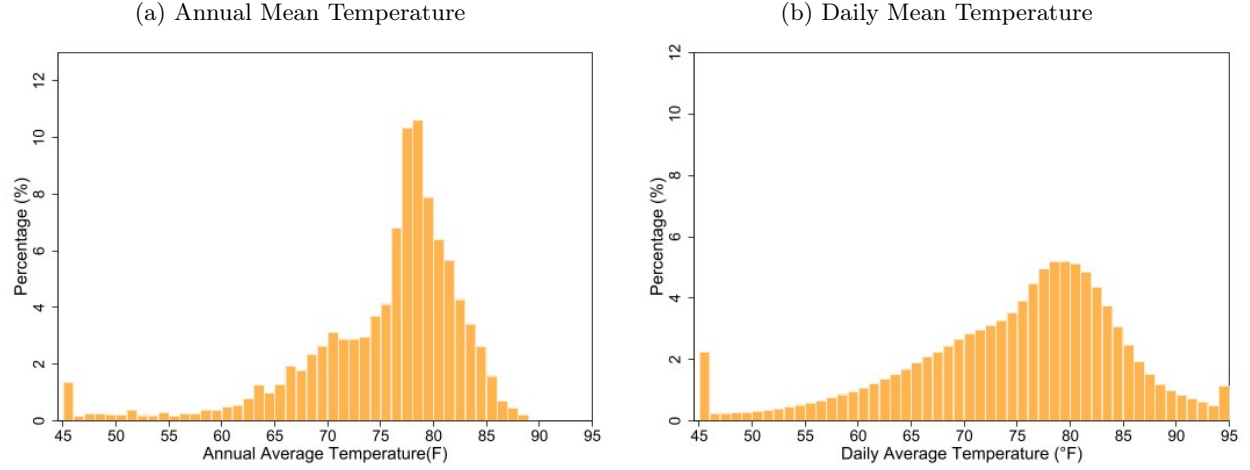


(b) Precipitation



Notes: These figure is based on ERA5 weather data for 2024. Panel (a) displays extreme temperature exposure. The red areas indicate regions that experience monthly temperatures above 90 °F at least once in the year, the blue areas indicate exposure to monthly temperatures below 50 °F, and the purple areas experience both extremes. Panel (b) shows extreme precipitation exposure. Yellow areas indicate regions with monthly precipitation below 5mm at least once in a year, blue areas show monthly precipitation above 300mm, and purple areas experience both drought and heavy rainfall conditions.

Figure 3: Distribution of Temperature Exposure Variables Across 57 Countries



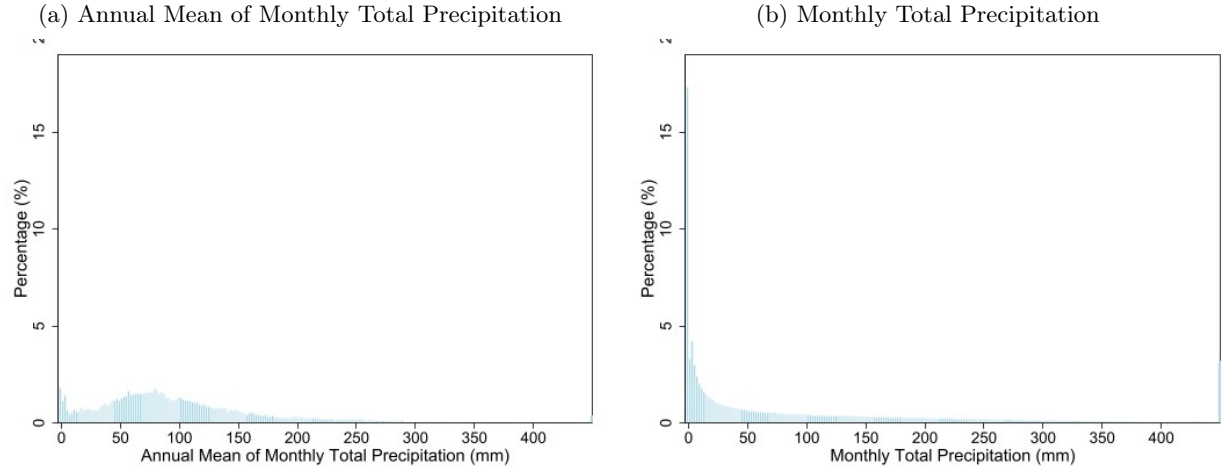
Notes: Panel (a) shows the distribution of children’s average temperature exposure over the 12 months prior to their health measurement, calculated by averaging all daily temperatures within that window. Panel (b) displays the frequency distribution of all daily temperature exposures experienced by children in our sample during the in utero and postnatal periods up to the time of health measurement.

aggregated to monthly totals (mm) to reflect cumulative rainfall. These measures enable mapping child-level nutrition outcomes to local weather over the full exposure window including prenatal and postnatal periods. This approach may introduce measurement error due to the spatial displacement of DHS GPS coordinates, although the ERA5 spatial resolution exceeds typical displacement distances and likely mitigates this concern. In addition, the prenatal period is inferred from age in months under a fixed gestation assumption, which may introduce imprecision in exposure timing.

Figures 3 and 4 summarize the distribution of weather exposure in the matched sample. Temperature exposure spans a wide range. While 77% of children experience annual mean temperatures between 70-85 °F, daily realizations include extremes above 90 °F and below 50 °F. Similarly, precipitation ranges from near-zero levels in arid regions to over 300 mm in tropical areas, with 49% of children exposed to moderate annual levels between 75 and 200 mm.

Extreme weather exposure is common in the matched sample. Daily temperature realizations capture exposure to both high and low extremes, with 42% of children experiencing temperatures above 90 °F and

Figure 4: Distribution of Precipitation Exposure Variables Across 57 Countries



Notes: Panel (a) shows the distribution of annual precipitation over the 12 months prior to each child’s health measurement, calculated by averaging monthly total precipitation within that window. Panel (b) displays the frequency distribution of all monthly precipitation exposures experienced by children in our sample during the in utero and postnatal periods up to the month of health measurement.

20% below 50 °F at least once between the prenatal period and the survey date. The monthly precipitation distribution indicates drought conditions (below 5 mm) for 83% of children and heavy rainfall exceeding 300 mm for 60%. These patterns provide variation in weather exposure across observations and form the basis for identifying nonlinear effects of weather on child nutrition. Many high-exposure areas are located in low- and middle-income countries, consistent with the global patterns shown in Figure 2.

3 Empirical Model

We estimate the causal effects of weather exposure across a child’s entire life cycle from conception to the survey date. For each child, we construct a complete history of weather exposure over this period. These exposure measures capture variation in both the intensity of weather realizations and the timing of exposure.

3.1 Baseline Specification

We employ restricted cubic spline(RCS) approach to flexibly capture nonlinear effects of weather on child stunting. (Hsiang, 2016; Deschênes and Greenstone, 2011). RCS uses piecewise cubic polynomials that are smoothly connected at predetermined knots. This allows the function to adapt flexibly to data patterns without imposing strong functional form assumptions (Harrell et al., 2001; Croxford, 2016).

An advantage of the RCS approach is its behavior at the boundaries. While high-order polynomials may exhibit extreme fluctuations and unstable extrapolation in the tails of the distribution, RCS imposes linearity beyond the boundary knots. This feature stabilizes the estimated response in sparsely populated regions. This is crucial for analyzing rare but extreme weather events.

We apply restricted cubic spline transformations to daily mean temperatures and monthly total precipitation over each child’s life cycle. For temperature, we place knots at 50°F, 65°F, 80°F, and 95°F. For precipitation, knots are positioned at 10mm, 100mm, 250mm, and 400mm.

We define the exposure window from conception to the survey date. For each child, we construct the full sequence of local temperature and precipitation exposures over this period. We then aggregate the transformed weather measures within three-month intervals over the life cycle. This aggregation yields a set of exposure measures that capture nonlinear weather realizations at different stages of early-life development. Because exposure is indexed by the timing of occurrence, the specification allows the effects of temperature and precipitation shocks to vary across gestation and early childhood.

We then estimate the following regression model:

$$Y_{igrmy,cz} = f(\text{Weather}_{igrmy,cz}) + \delta_{cy} + \nu_{rm,cz} + \epsilon_{igrmy,cz}$$

where $Y_{igrmy,cz}$ denotes the child outcome (stunting z-score) for child i residing in grid cell g , province r , country c , interviewed in month m of year y , in climate zone cz .

$f(\cdot)$ represents the restricted cubic spline functions of temperature and precipitation exposures. We implement these exposures using a distributed lag structure defined over three-month intervals. For each age

group, the specification includes a contemporaneous term(q_0) and lag terms (q_1 through q_j) that extend back through the prenatal period, with each lag capturing exposure over a three-month period.

The number of lags varies with the child’s age at the time of measurement. And we estimate the model separately for each age group. The 6-11 month group includes five lags, covering 15 months (nine prenatal months and six months after birth). the 12-23 month group includes seven lags(21 months), the 24-35 month group includes 11 lags (33 months), and the 36-59 month group includes 15 lags (45 months). In all specifications, the three earliest lag terms q_j q_{j-2} correspond to exposure during pregnancy, capturing the first, second, and third trimesters. The remaining lags capture the postnatal exposure. The contemporaneous term (q_0) captures residual exposure that does not fall within a complete three-month interval. This covers the period between the survey month and q_1 . We also include two quarterly lead terms (q_{-1}, q_{-2}) for placebo tests.

Our fixed effects isolate plausibly exogenous variation in children’s weather exposure histories over the life cycle from conception to the survey date. First, we include country-year fixed effects δ_{cy} to absorb national-level variation that evolves annually, including changes in macroeconomic conditions, food policy, health interventions, and other institutional factors. Second, we include province-climate zone-month fixed effects $\nu_{rm,cz}$, which absorb average seasonal patterns in child outcomes within each province-climate zone.² The remaining identifying variation reflects idiosyncratic deviations in temperature and precipitation from typical monthly patterns within each province-climate zone over the child’s life cycle, from conception to the survey date, after accounting for country-year shocks. Further, the same weather shock can affect children at different ages. This allows us to identify how the weather effects vary across stages of early-life development.

We cluster standard errors at the country-year level to account for arbitrary correlation in unobservables within countries over time.

²Climate zone refers to Köppen-Geiger climate classifications with 30 distinct characteristic temperature and precipitation regimes over the globe (Kottek et al., 2006).

3.2 Heterogeneity Analysis

Weather shocks may vary across demographic groups. We estimate heterogeneity by interacting weather exposure with demographic indicators in the base line specification:

$$Y_{igrmcy,cz} = f(\text{Weather}_{igrmcy,cz}) \times D_i + \alpha \cdot D_i + \delta_{cy} + \nu_{rm,cz} + \epsilon_{igrmcy,cz}$$

Here, D_i denotes a demographic indicator for child i . The baseline function $f(\cdot)$ captures the weather-stunting relationship for the reference group ($D_i = 0$), while the interaction term captures differential responses for $D_i = 1$. We estimate these specifications using the same restricted cubic spline approach as in the baseline model. We focus in particular on the combined effect of child sex and household wealth.

4 Results and Discussion

We estimate nonlinear effects of temperature and precipitation on child height-for-age-z scores (HAZ) over the period from conception to the survey date. Effects are normalized relative to moderate reference conditions of 75°F for temperature and 175mm for monthly precipitation, so each estimate can be interpreted as the effect of deviations from these baselines. For temperature, the figures report estimated effects per 10 exposure days to a given daily mean temperature. This scaling is for interpretability and does not refer to 10 consecutive days. We focus on children aged 36-59 months because, for this group, exposure over the first 1000 days is fully observed and observed height is more likely to reflect persistent growth deficits. Among children aged 24-35 months, catch-up growth may still be relatively common, so measured height may be a noisier indicator of lasting early-life damage. Results for younger age groups are reported in the Appendix.

4.1 Temperature

Figure 5 shows the estimated marginal effects of temperature exposure on HAZ for children ages 36–59 months. The x-axis gives daily average temperature in Fahrenheit, and the y-axis shows the estimated effect of 10 exposure days on HAZ. Negative estimates indicate lower height-for-age relative to the reference

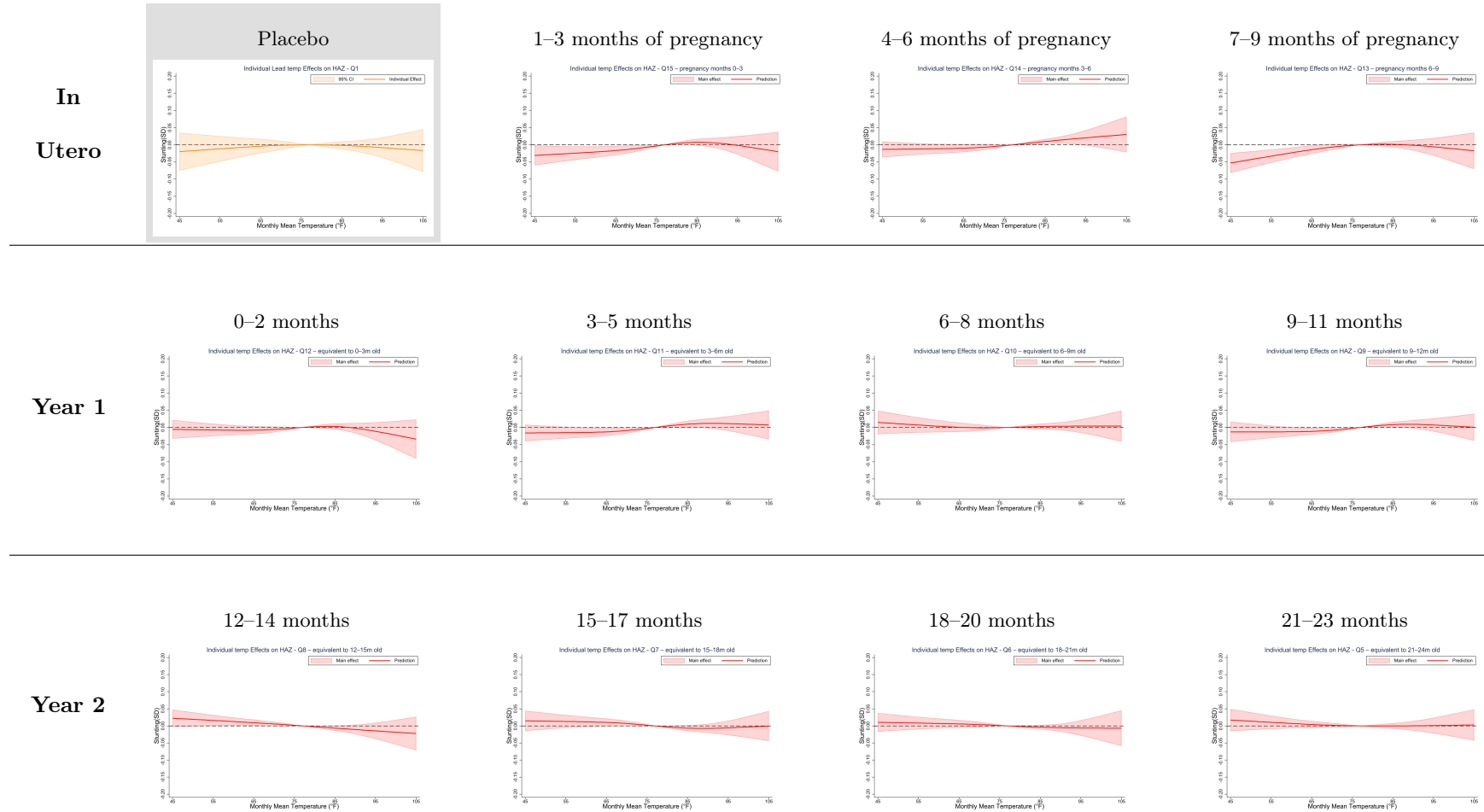
condition, implying slower child growth.

The main pattern is that cold exposure in utero lowers HAZ, with the largest effects concentrated in late pregnancy. The relationship is nonlinear and HAZ declines more sharply as temperatures fall further below moderate conditions. The strongest effects appear during months 7–9 of pregnancy. For example, 10 exposure days at daily mean temperature below 45 °F during the third trimester reduce HAZ by 0.05 standard deviations relative to the 75 °F reference level.

The estimates compare children with different early-life temperature realizations after absorbing country-year and province-climate zone-month fixed effects. The remaining variation reflects idiosyncratic deviations in local temperature relative to typical monthly patterns within the same province-climate zone after accounting for country-year shocks. The pattern therefore suggests that unusually cold conditions during gestation, especially late pregnancy, causally reduce child height growth (lower HAZ) and that these effects remain evident after age 36 months. One interpretation is that unusually cold conditions during gestation impair prenatal development through maternal health and nutritional stress, with especially important consequences late in pregnancy when the growth is most rapid. The estimates are also consistent with broader household channels operating through agricultural production, labor income, and the household environment. In particular, prenatal cold exposure may also affect agricultural output in the relevant period. Our reduced-form design does not separately identify these mechanisms.

Postnatal temperature exposure has weaker and less precisely estimated effects than prenatal exposure. During the first year of life, we find some evidence that cold exposure at ages 3–5 months and 9–11 months lowers HAZ, although the estimates become noisier as temperatures become more extreme. The estimated effects are 0.015 standard deviations per 10 exposure days. Relative to reference weather, slightly cooler temperatures lower HAZ, while moderately warmer temperatures raise HAZ. In the second year of life, the negative temperature effects become less precisely estimated relative to the prenatal period. Exposure to slightly cooler temperatures around ages 12–14 months raises HAZ relative to the reference temperature. The clearest negative effects appear during gestation, especially in late pregnancy.

Figure 5: Temperature Effects on HAZ (Age Group: 36–59 Months Old)



Notes:

4.2 Precipitation

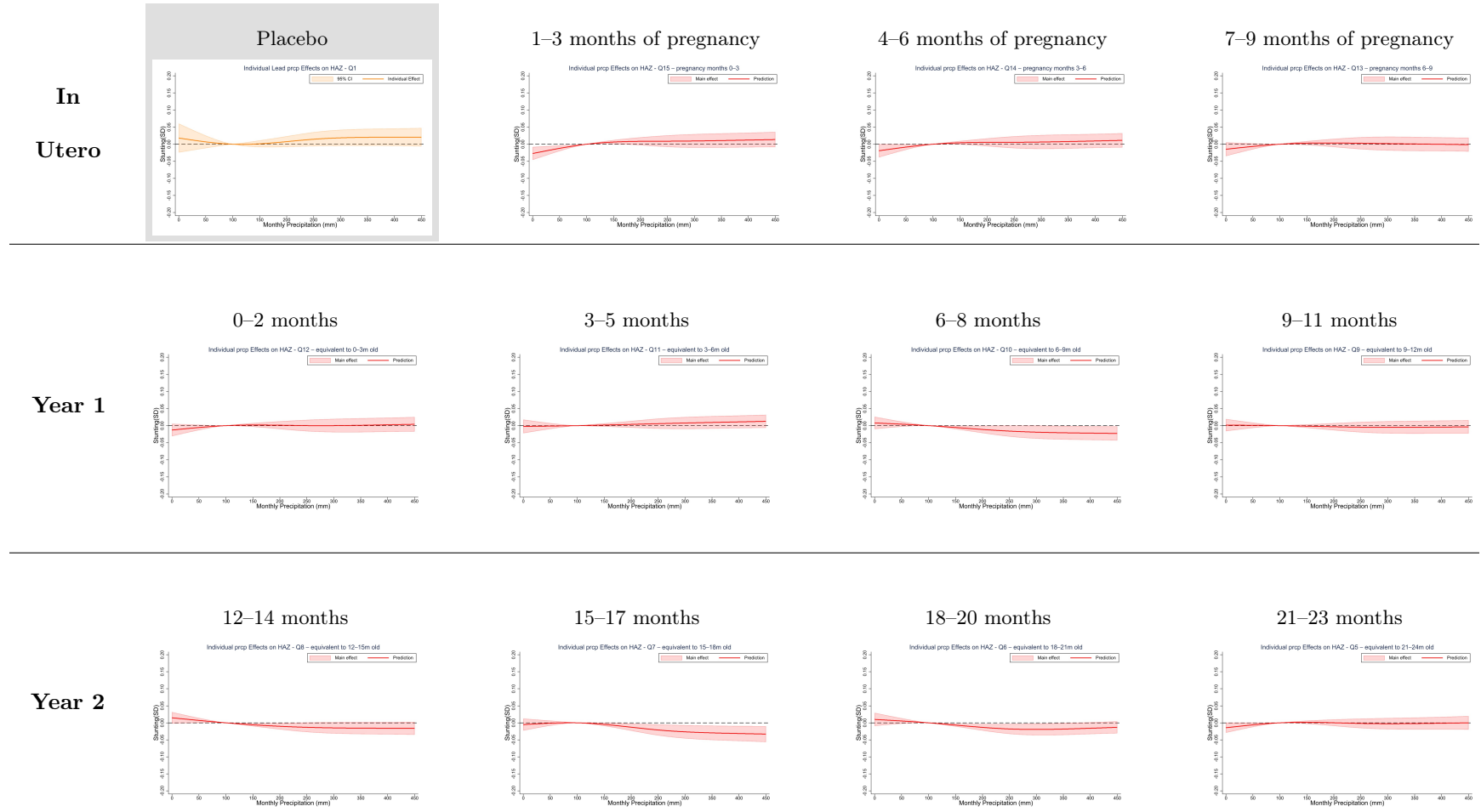
Figure 6 shows the estimated marginal effects of precipitation exposure on HAZ for children ages 36–59 months. The x-axis gives monthly total precipitation in millimeters, and the y-axis shows the estimated effect of an additional month of exposure on HAZ. Negative estimates indicate lower height-for-age relative to the reference condition.

The main pattern is that drought exposure during gestation lowers HAZ, with the strongest negative effects appearing during months 1–3 pregnancy. For example, exposure to very low precipitation during the first trimester lowers HAZ by about 0.027 standard deviations relative to the reference precipitation condition.

The estimates compare children exposed to different local precipitation realizations after accounting for country conditions that change over time and seasonal patterns within the same province-climate zone. The remaining variation comes from unusual local precipitation relative to typical monthly conditions in that same area. The pattern therefore indicates that unusually low precipitation during gestation causally lowers height-for-age. A potential mechanism is that drought exposure before birth contribute to growth faltering through maternal health and nutritional stress. This pattern does not rule out broader agricultural channels operating through production, income, and the household environment. Our empirical design does not separately identify the channels underlying these effects.

Postnatal precipitation exposure follows a different pattern. We find little evidence that precipitation affects HAZ during the early part of the first year of life. Negative effects of heavy rainfall begin to appear at ages 6–8 months, when an additional month of exposure lowers HAZ by 0.023 standard deviations. In the second year, this pattern recurs at ages 15–17 months, when heavy rainfall decreases HAZ by 0.033 standard deviations. Negative effects of heavy rainfall are also visible in other quarters of the second year, although the estimates are less precise. Overall, the precipitation results indicate that drought matters most before birth, while heavy rainfall becomes more relevant after birth.

Figure 6: Precipitation Effects on HAZ (Age Group: 36–59 Months Old)



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4.3 Heterogeneity by Gender and Household Wealth

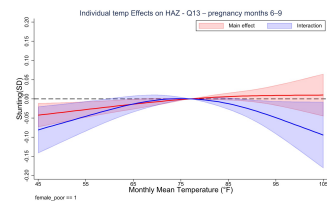
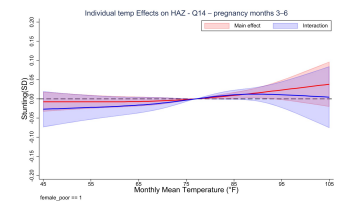
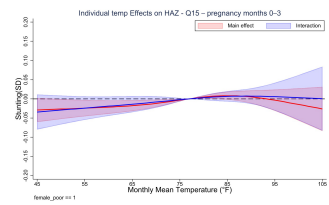
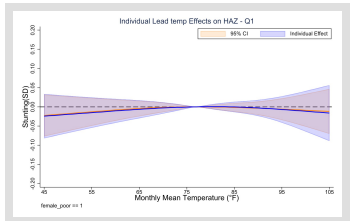
This subsection examines heterogeneity by child sex and household wealth. Overall, girls in the bottom two wealth quintiles appears more vulnerable to weather extremes than other children. In figure 7, this pattern is most visible during prenatal period, especially in the third trimester, when the decline in HAZ is larger for poor girls. Figure 8 shows a similar vulnerability pattern for precipitation, although the estimates are imprecise and some lead effects are present.

These results should be interpreted with caution. We do not find clear evidence that the estimated responses differ statistically across groups. Appendix A reports heterogeneity by sex alone and by household wealth alone.

Appendix A reports additional heterogeneity analyses by rural residence and by parental involvement in agriculture, measured by whether either parent works in the agricultural sector. These comparisons are economically relevant because exposure to weather shocks and the capacity to mitigate them may vary systematically across settings. In practice, however, the estimates are noisy and not statistically distinguishable across groups.

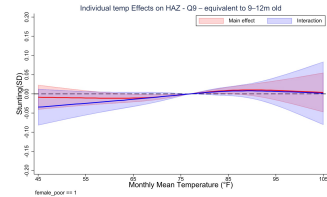
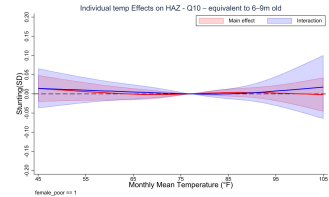
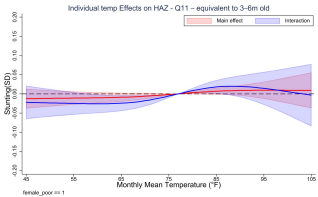
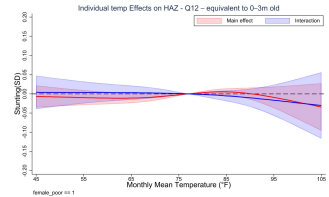
Figure 7: Temperature Effects: Girls in the Bottom Two Wealth Quintiles

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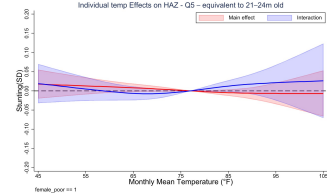
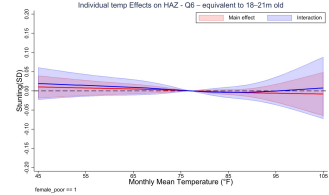
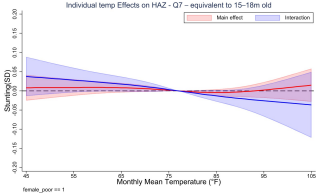
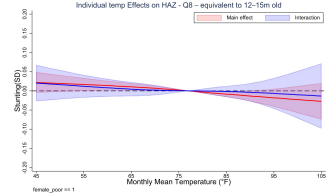


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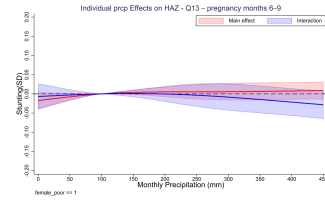
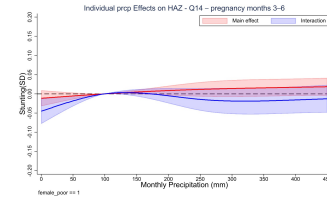
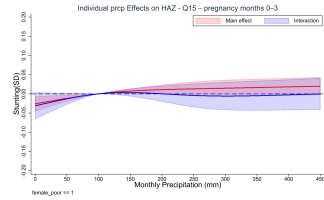
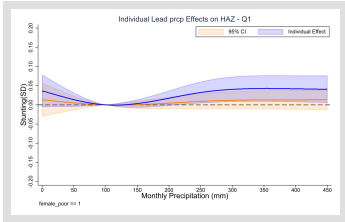
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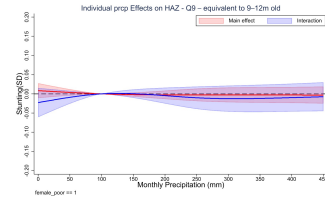
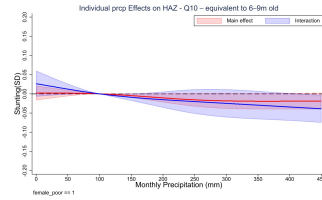
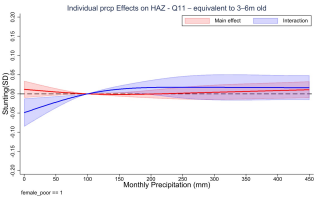
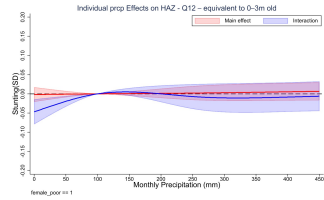
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Figure 8: Precipitation Effects: Girls in the Bottom Two Wealth Quintiles

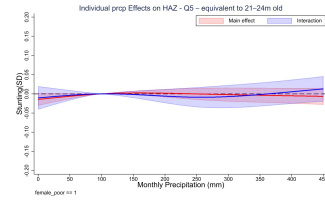
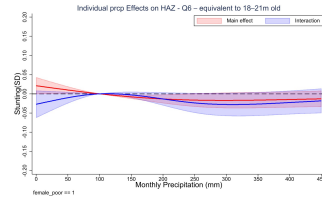
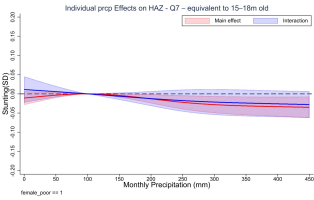
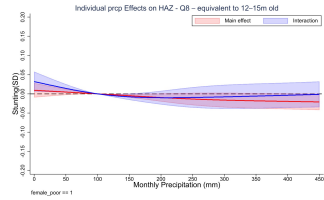
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5 Conclusion

Using micro-level data from a large set of low- and middle-income countries and complete exposure histories over early life, we document persistent effects of weather exposure on child health and growth. The results show that these effects vary by both the timing of exposure and the type of shock. For children observed at ages 36 to 59 months, cold exposure during gestation, especially in late pregnancy, has persistent negative effects on height-for-age(HAZ). Precipitation follows a different timing pattern, with drought during early pregnancy and heavy rainfall after birth each lowering HAZ. Taken together, adverse weather affects child growth through distinct periods of vulnerability rather than through a single early-life window.

These patterns are consistent with multiple channels, but our reduced-form design does not separately identify them. The gestational effects of cold and drought are consistent with maternal health and nutritional stress during pregnancy. The postnatal effects of heavy rainfall are consistent with channels that become more relevant after birth, including disease exposure, sanitation disruptions, and other shocks to the household environment. Broader household responses may also matter, since weather shocks can affect agricultural production, household income, food access, and caregiving conditions.

These findings matter because the effects remain visible among children ages 36 to 59 months, when deficits in height are less likely to be fully reversed through catch-up growth, absent sufficient intervention, and are therefore more likely to represent persistent disadvantage rather than only short-run nutritional deficits. This is important in a development context because deficits that remain at older ages are more likely to carry forward into later human capital outcomes. Our results therefore suggest that weather shocks over early life may shape disadvantage well beyond the period of immediate exposure.

Several limitations remain. First, the analysis matches child-level outcomes to ERA5 weather data using the child’s age in months and the survey month, which may introduce measurement error because exact dates are not used. Second, the analysis identifies effects only among children who survive to the survey date. If extreme weather also causes severe health shocks that increase mortality risk, our estimates will miss those most severe negative effects and may therefore understate the full impact of weather exposure. Third, we

do not directly observe adaptation responses to weather shocks. While adaptive capacity may be limited in many low- and middle-income settings, we do not measure these margins directly. Future work could examine these margins more explicitly.

The policy relevance of these findings extends beyond short-run weather fluctuations. As climate variability increases and climate change alters the distribution of adverse weather conditions, the risks documented here may become increasingly important for maternal and child health in vulnerable populations. The findings also point to the importance of protective capacity during pregnancy and early childhood, including access to nutrition, health care, and safer household environments. More broadly, they suggest that climate-related risks to development operate in part through early-life human capital formation, making maternal and child protection a relevant dimension of climate adaptation policy.

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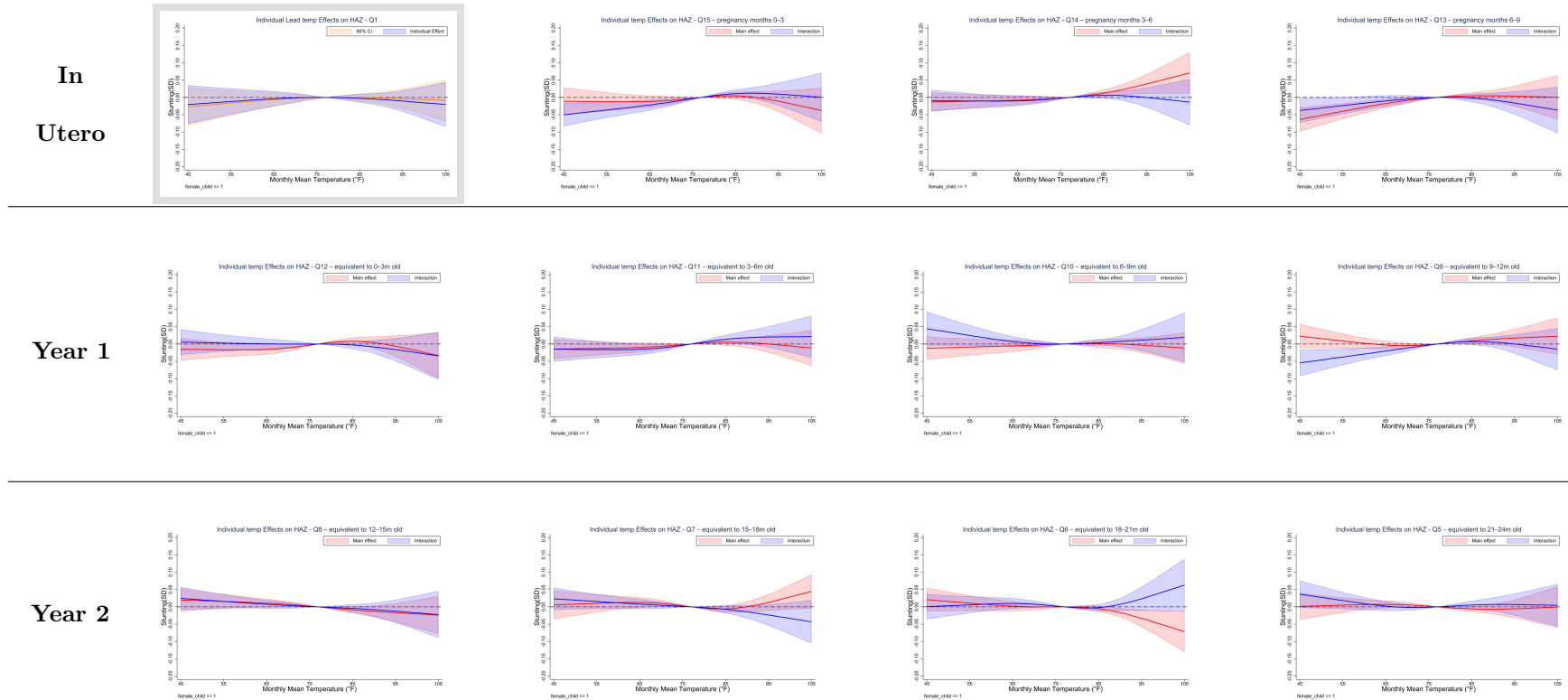
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Appendix A Additional Heterogeneity Checks

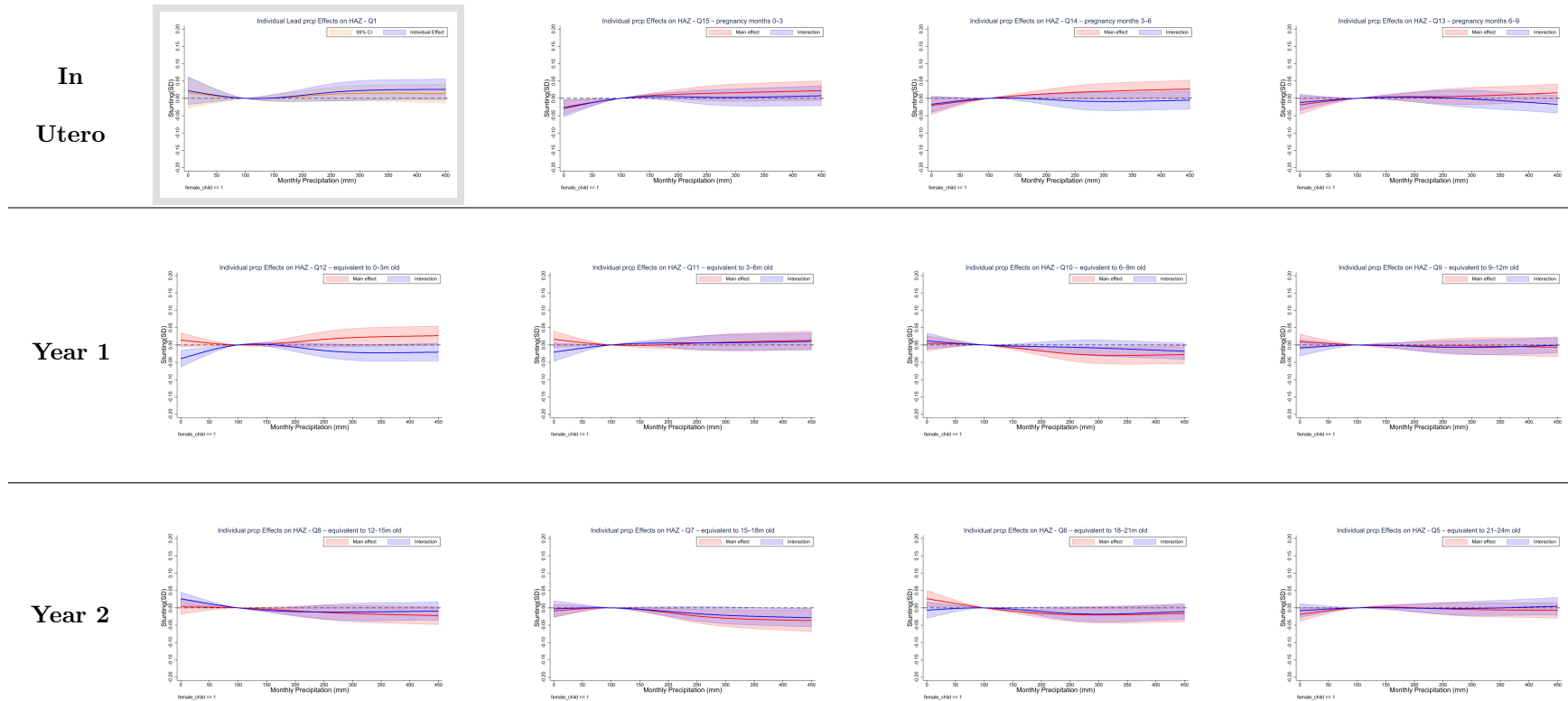
A.0.1 Heterogeneity by Sex

Figure 9: Temperature Effects: Heterogeneity by Sex



Notes: The figure reports the estimated marginal effects of temperature exposure on height-for-age z-scores (HAZ) among children aged 36-59 months. The estimates come from specifications that interact weather exposure with an indicator for female children. The effects are reported separately for boys, the reference group, and girls. Temperature exposure is defined as the number of days within each three-month interval at a given daily mean temperature level, and all effects are scaled to a ten-day increase.

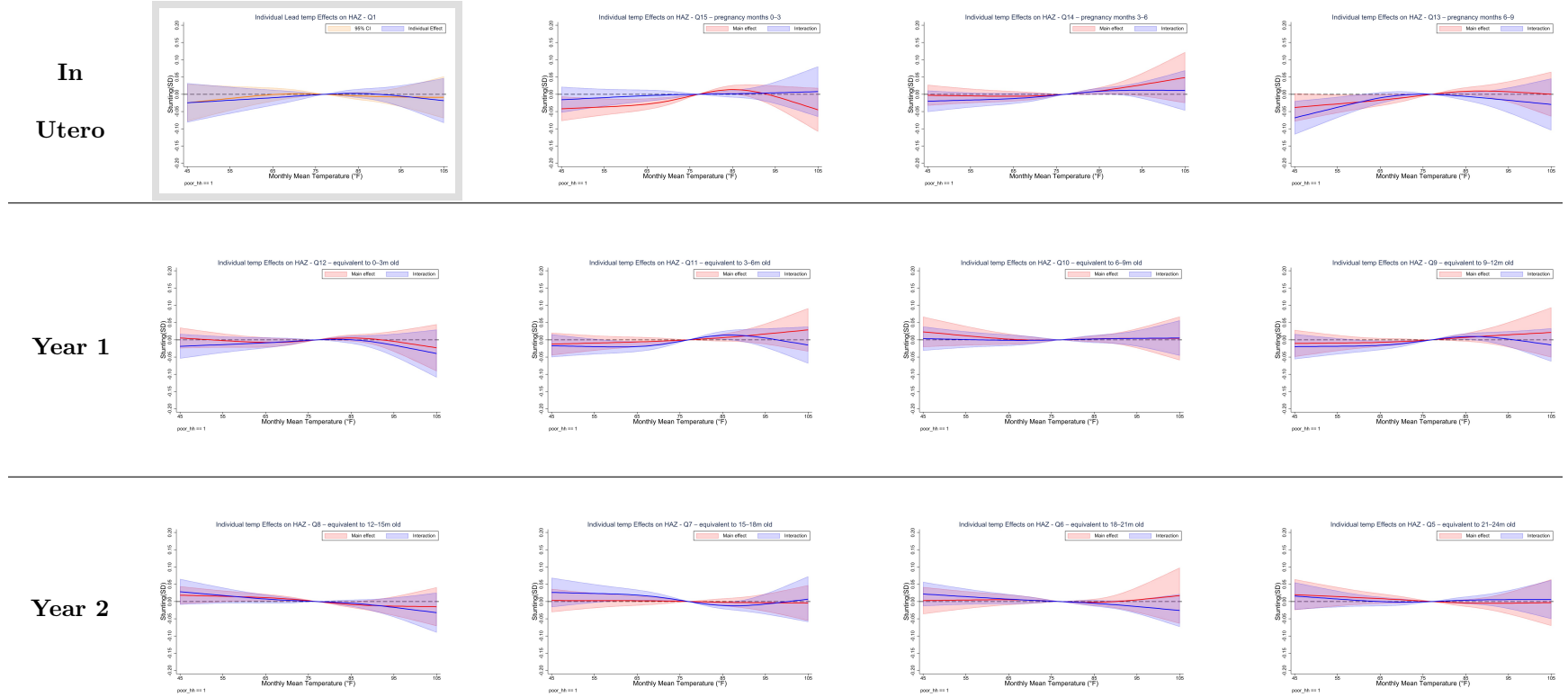
Figure 10: Precipitation Effects: Heterogeneity by Sex



Notes: The figure reports the estimated marginal effects of precipitation exposure on height-for-age z-scores (HAZ) among children aged 36-59 months. The estimates come from specifications that interact weather exposure with an indicator for female children. The effects are reported separately for boys, the omitted group, and girls. Precipitation exposure is measured as monthly total rainfall, in millimeters, within each three-month interval.

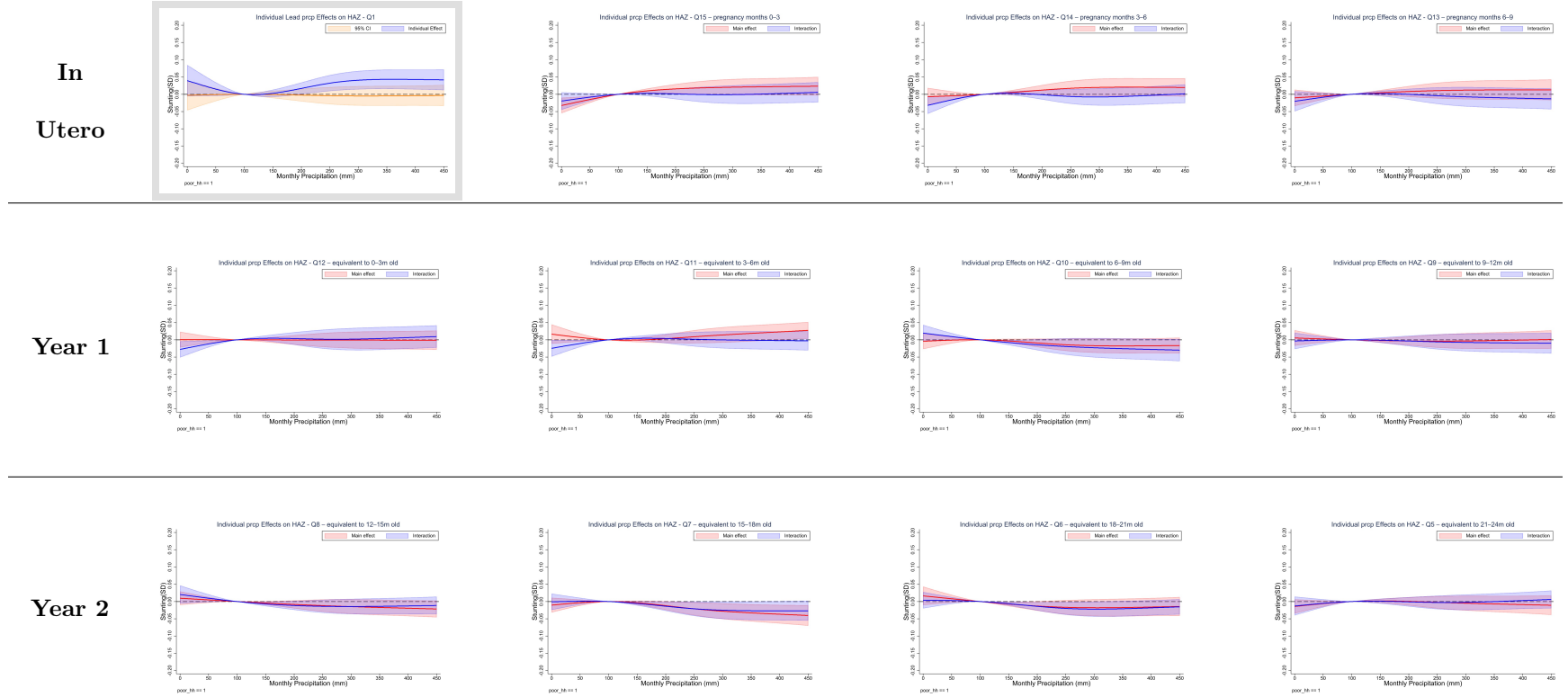
A.0.2 Heterogeneity by Income Level

Figure 11: Temperature Effects: Heterogeneity by Income Level



Notes: The figure reports the estimated marginal effects of temperature exposure on height-for-age z-scores (HAZ) among children aged 36-59 months. The estimates come from specifications that interact weather exposure with an indicator equal to one for children in the bottom two income quintiles. The effects are reported separately for children in the top three income quintiles, the reference group, and those in the bottom two income quintiles. Temperature exposure is measured as the number of days within each three-month interval at a given daily temperature level, and all effects are scaled to a ten-day increase.

Figure 12: Precipitation Effects: Heterogeneity by Income Level

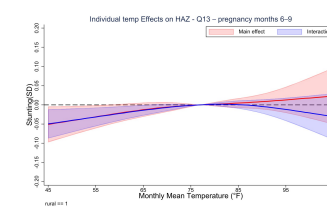
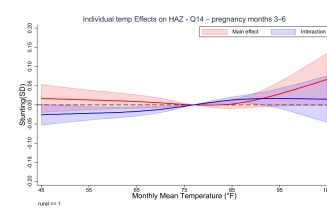
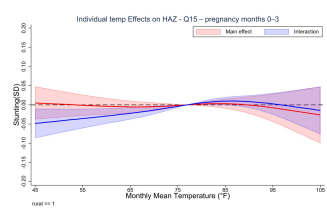
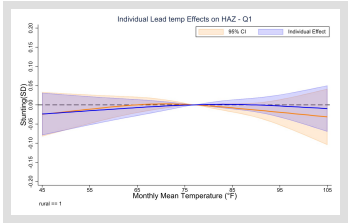


Notes: The figure reports the estimated marginal effects of precipitation exposure on height-for-age z-scores (HAZ) among children aged 36-59 months. The estimates come from specifications that interact weather exposure with an indicator equal to one for children in the bottom two income quintiles. The effects are reported separately for children in the top three income quintiles, the omitted group, and those in the bottom two income quintiles. Precipitation exposure is measured as monthly total rainfall, in millimeters, within each three-month interval.

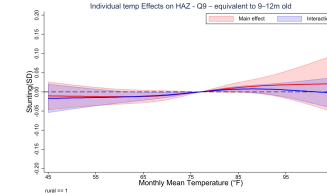
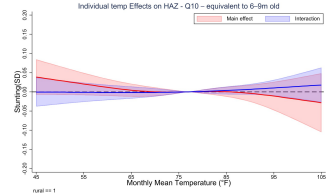
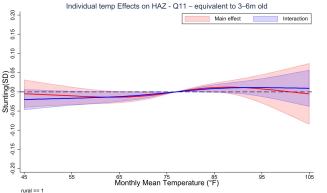
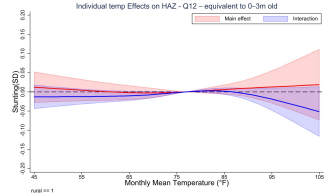
A.0.3 Urban-Rural Differences in Weather Effects

Figure 13: Temperature Impact Heterogeneity: Rural–Urban Differences

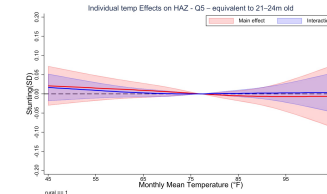
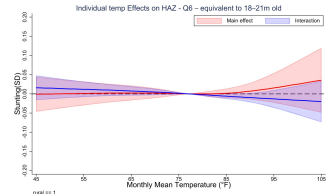
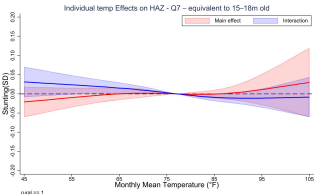
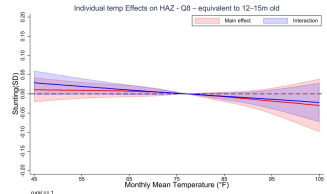
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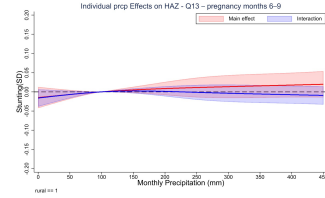
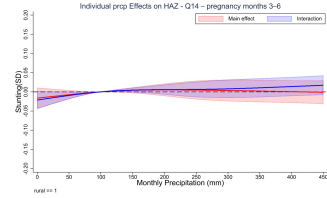
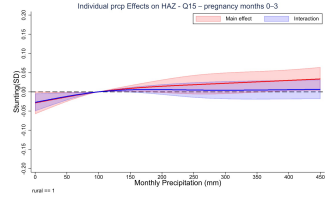
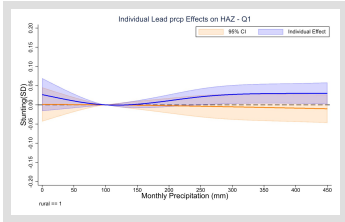
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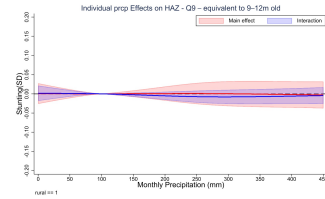
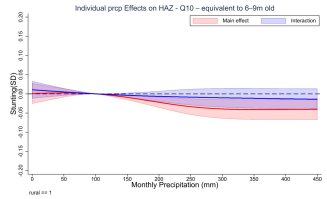
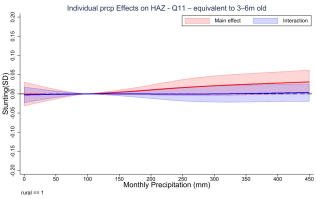
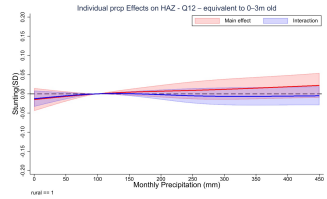
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Figure 14: Precipitation Impact Heterogeneity: Rural–Urban Differences

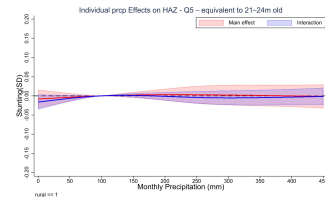
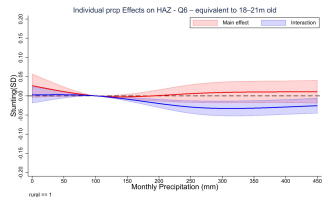
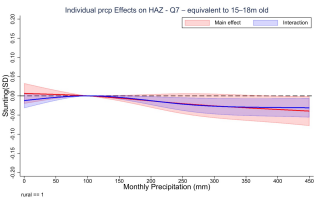
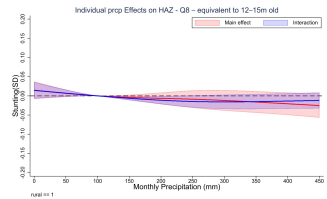
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